

Lab 10: Structural Bioinformatics

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PDB statistics

```
pdb_stats <- read.csv("pdb_stats.csv")
head(pdb_stats)
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	178795	21825	12773	343	226
2	Protein/Oligosaccharide	10363	3564	34	8	11
3	Protein/NA	9106	6335	287	24	7
4	Nucleic acid (only)	3132	221	1566	3	15
5	Other	175	25	33	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron Other Total					
1	84 32	214078				
2	1 0	13981				
3	0 0	15759				
4	3 1	4941				
5	0 0	237				
6	0 4	22				

Q1: What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
total <- 253482
xray <- 203939
em <- 34005

((xray + em) / total) * 100
```

```
[1] 93.87018
```

About 93.87% of PDB structures are solved by X-ray or Electron Microscopy.

Q2: What proportion of structures in the PDB are protein?

```
total <- 253482

protein_only <- 180891 + 23208 + 12816 + 350 + 229 + 84
protein_oligo <- 10495 + 3745 + 34 + 8 + 11 + 1
protein_na <- 9206 + 6773 + 287 + 26 + 8 + 0

protein_total <- protein_only + protein_oligo + protein_na

protein_total / total
```

```
[1] 0.9790518
```

```
(protein_total / total) * 100
```

```
[1] 97.90518
```

The proportion of PDB structures that are protein is about 0.979, or 97.91%

Q3: Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are currently 1,227 HIV-1 protease structures in the PDB.

The important role of water

Q4: Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

Water molecules normally have 3 atoms, but in X-ray crystal structures the hydrogen atoms are usually not visible because they scatter X-rays very weakly. Because of this, each water molecule is usually shown as only the oxygen atom.

Q5: There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

The conserved water molecule in the binding site is water molecule HOH 308. This water molecule is located between the ligand and the flap residues near Ile-50 in the active site and helps stabilize ligand binding through hydrogen bonding interactions.

Q6



Figure 1: HIV-1 protease (1HSG) showing two chains, indinavir ligand, ASP25 residues and conserved water HOH308

Introduction to Bio3D in R

```
library(bio3d)
```

```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
pdb
```

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
```

```
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD  
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE  
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP  
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
       calpha, remark, call
```

Q7: How many amino acid residues are there in this pdb object?

198 residues

Q8: Name one of the two non-protein residues?

HOH

Q9: How many protein chains are in this structure?

2 chains

```
attributes(pdb)
```

```
$names
```

```
[1] "atom" "xyz" "seqres" "helix" "sheet" "calpha" "remark" "call"
```

```
$class
```

```
[1] "pdb" "sse"
```

```
head(pdb$atom)
```

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40

	segid	elesy	charge
1	<NA>	N	<NA>
2	<NA>	C	<NA>
3	<NA>	C	<NA>
4	<NA>	O	<NA>
5	<NA>	C	<NA>
6	<NA>	C	<NA>

```
library(bio3dview)
```

```
library(NGLViewerR)
```

```
#view.pdb(pdb) |>  
#setSpin()
```

```
# Select the important ASP 25 residue
```

```
sele <- atom.select(pdb, resno=25)
```

```
# and highlight them in spacefill representation
```

```
#view.pdb(pdb, cols=c("navy","teal"),  
          #highlight = sele,  
          #highlight.style = "spacefill") |>  
#setRock()
```

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
```

```
Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)
```

```
Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 244 (residues: 244)
```

```
Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

```
Protein sequence:
```

```
MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLV  
DELVIALVKERIAQEDCRNGFLLDGFPRTIPQADAMKEAGINVDYVLEFDVPDELIVDKI  
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG  
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,  
      calpha, remark, call
```

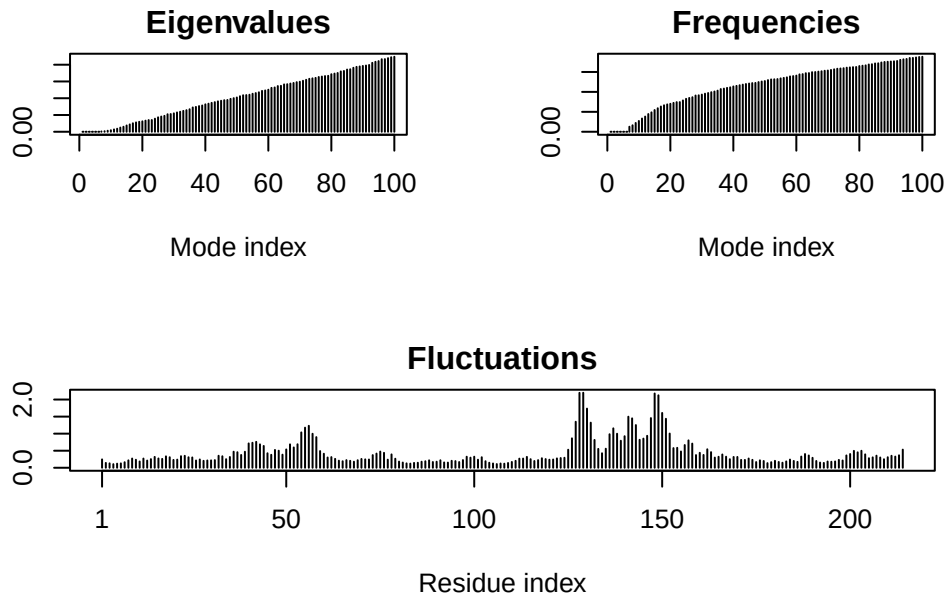
```
# Perform flexibility prediction
```

```
m <- nma(adk)
```

```
Building Hessian... Done in 0.02 seconds.
```

```
Diagonalizing Hessian... Done in 0.461 seconds.
```

```
plot(m)
```



```
#mktrj(m, file="adk_m7.pdb")
#view.nma(m, pdb=adk)
```

Comparative structure analysis of Adenylate Kinase

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa.

Q11. Which of the above packages is not found on BioConductor or CRAN?:

bio3dview.

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True.

```
library(bio3d)

# Get the ADK sequence
aa <- get.seq("1ake_A")
```

Warning in get.seq("lake_A"): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

aa

```

      1      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMLRAAVKSGSELGKQAKDIMDAGKLVT
      1      .      .      .      .      .      60

      61      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRPTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      120

     121      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKFNPVKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
     121      .      .      .      .      .      180

     181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
     181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

+ attr: id, ali, call

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

214 amino acids

```
hits <- NULL
```

```
hits$ pdb.id <- c('1AKE_A','6S36_A','6RZE_A','3HPR_A','1E4V_A','5EJE_A','1E4Y_A','3X2S_A','6H
```



```
files <- get.pdb(hits$pdb.id, path="pdbs", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6S36.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6RZE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/3HPR.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/1E4V.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/5EJE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/1E4Y.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/3X2S.pdb.gz exists. Skipping download
```

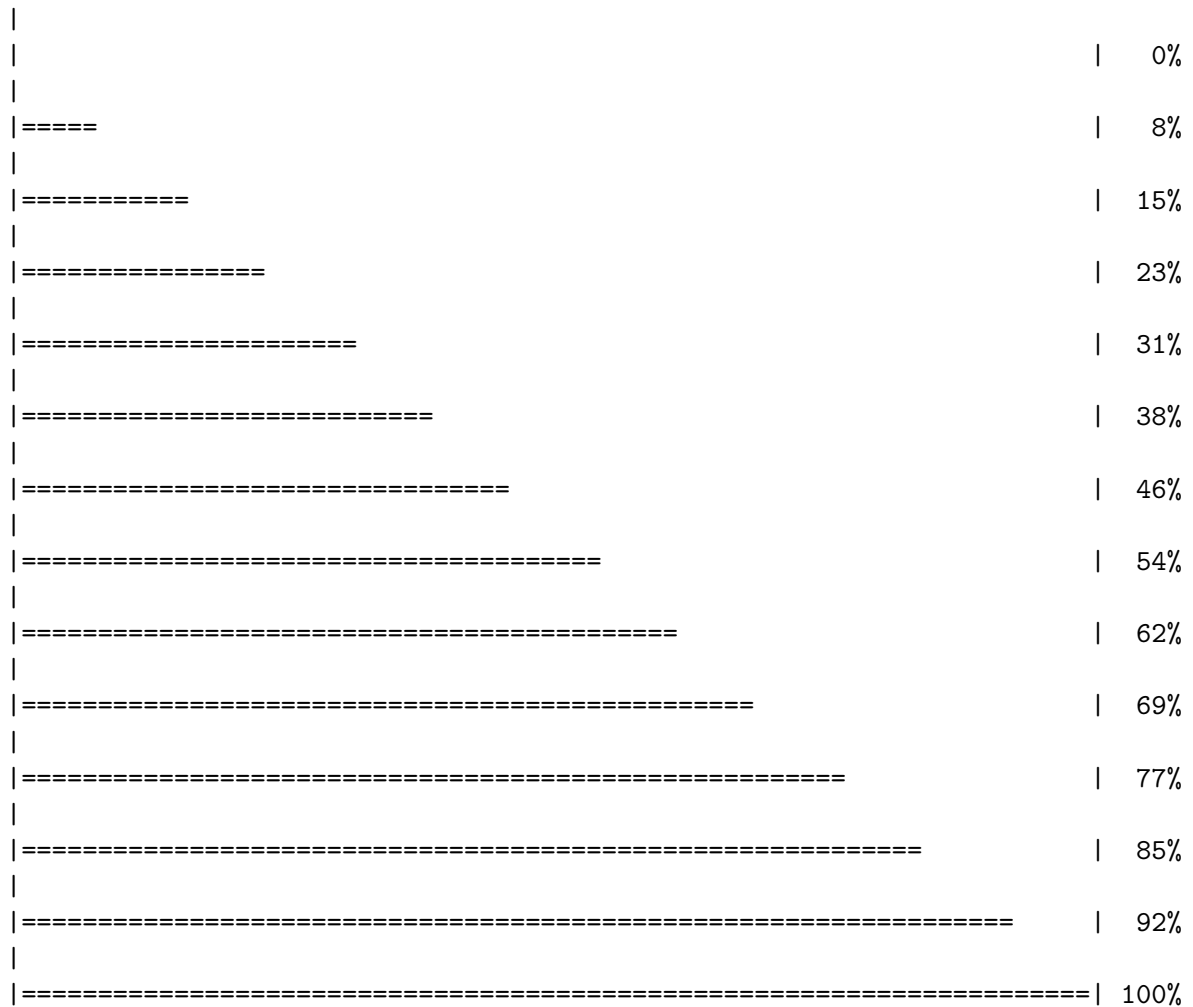
```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6HAP.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/6HAM.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/4K46.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):  
pdbs/3GMT.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4PZL.pdb.gz exists. Skipping download
```



```
pdbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```
pdbs/split_chain/1AKE_A.pdb
pdbs/split_chain/6S36_A.pdb
pdbs/split_chain/6RZE_A.pdb
pdbs/split_chain/3HPR_A.pdb
pdbs/split_chain/1E4V_A.pdb
pdbs/split_chain/5EJE_A.pdb
```

```

pdb/split_chain/1E4Y_A.pdb
pdb/split_chain/3X2S_A.pdb
pdb/split_chain/6HAP_A.pdb
pdb/split_chain/6HAM_A.pdb
pdb/split_chain/4K46_A.pdb
pdb/split_chain/3GMT_A.pdb
pdb/split_chain/4PZL_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
...

```

Extracting sequences

```

pdb/seq: 1   name: pdb/split_chain/1AKE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdb/split_chain/6S36_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdb/split_chain/6RZE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 4   name: pdb/split_chain/3HPR_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdb/split_chain/1E4V_A.pdb
pdb/seq: 6   name: pdb/split_chain/5EJE_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdb/split_chain/1E4Y_A.pdb
pdb/seq: 8   name: pdb/split_chain/3X2S_A.pdb
pdb/seq: 9   name: pdb/split_chain/6HAP_A.pdb
pdb/seq: 10  name: pdb/split_chain/6HAM_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 11  name: pdb/split_chain/4K46_A.pdb
    PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdb/split_chain/3GMT_A.pdb
pdb/seq: 13  name: pdb/split_chain/4PZL_A.pdb

```

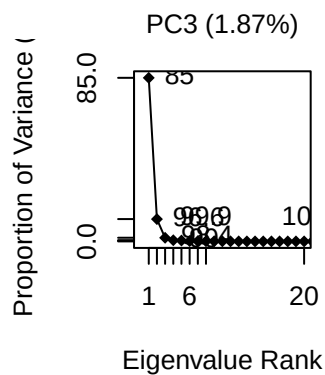
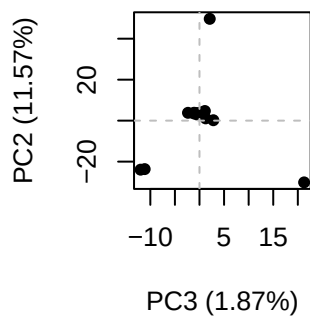
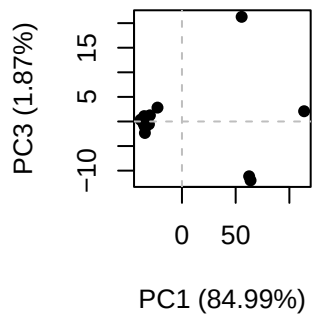
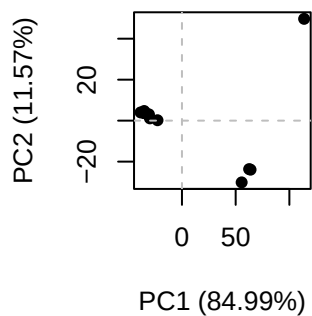
Annotate collected PDB structures

```
# Vector containing PDB database codes
ids <- basename(pdb(pdb$id))

anno <- pdb.annotate(ids)
unique(anno$source)
```

```
[1] "Escherichia coli"
[2] "Escherichia coli K-12"
[3] "Escherichia coli O139:H28 str. E24377A"
[4] "Escherichia coli str. K-12 substr. MDS42"
[5] "Photobacterium profundum"
[6] "Burkholderia pseudomallei 1710b"
[7] "Francisella tularensis subsp. tularensis SCHU S4"
```

```
# Perform PCA
pc.xray <- pca(pdb)
plot(pc.xray)
```

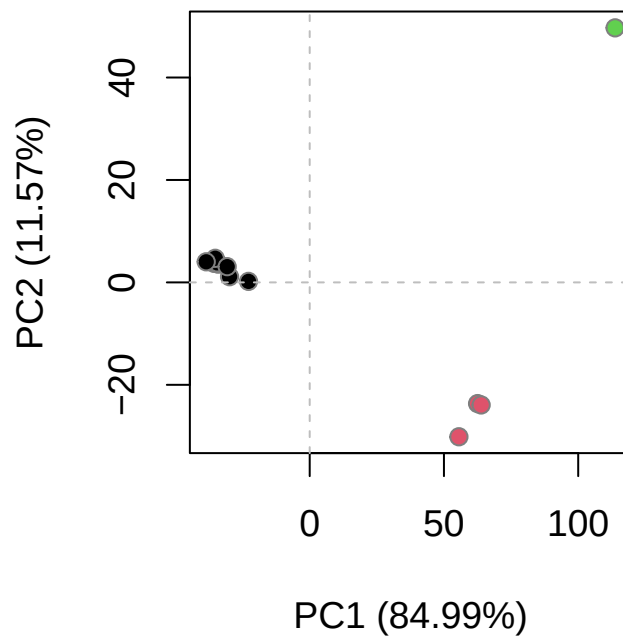


```
# Calculate RMSD
rd <- rmsd(pdbbs)
```

Warning in rmsd(pdbbs): No indices provided, using the 204 non NA positions

```
# Structure-based clustering
hc.rd <- hclust(dist(rd))
grps.rd <- cutree(hc.rd, k=3)

plot(pc.xray, 1:2, col="grey50", bg=grps.rd, pch=21, cex=1)
```



```
# Visualize first principal component
pc1 <- mktrj(pc.xray, pc=1, file="pc_1.pdb")
```

```
#Plotting results with ggplot2
library(ggplot2)
library(ggrepel)

df <- data.frame(PC1=pc.xray$z[,1],
                  PC2=pc.xray$z[,2],
                  col=as.factor(grps.rd),
```

```

ids=ids)

p <- ggplot(df) +
  aes(PC1, PC2, col=col, label=ids) +
  geom_point(size=2) +
  geom_text_repel(max.overlaps = 20) +
  theme(legend.position = "none")
p

```

